CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CL)

MAY 2014

CL 208 Advance Surveying

Date: 03.05.2014, Saturday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q-1 To determine the elevation of a water tank, a theodolite was kept two stations A and B 300m apart, A being nearer to the tank. The readings at the benchmark (of R.L. 1030) were 1.30m from station A and 2.10m from station B. The vertical angles to the top of the tank were 20° 30' and 7° 15'from station A and B respectively. Find the horizontal Distance and R.L. on top of the water tank. **[07]**

Q-2 Attempt Any Two **[14]**

Q-2(A) **i) Find multiplying and additive Constant form the following observation.**

Instrument at	Observation to	Distance	Staff reading
A	B	30m	1.135,1.285,1.435
A	C	60m	1.025,1.324,1.625

ii) An instrument was setup at 25 m away from transmission tower. The angle of elevation to the top of the tower was 20° 15' and the angle of depression to the bottom was 5°. Calculate height of the tower.

Q-2(B) For tachometer with and analytic lens following observation were taken. Calculate distance AB and gradient of line AB. Staff reading on B.M=2.1m, (R.L of B.M 150m) and K=100

Inst. St.	Staff reading on B.M	Staff St.	WCB	Vertical angle	Hair reading
O	1.55	A	40°30'	4°30'	1.150, 1.750, 2.350
O	1.55	B	85°30'	10°15'	1.250, 2.000, 2.750

Q-2(C) For performing tacheometry survey with theodolite having only two staida hair one is vertical and another is horizontal. Kepping one angle in elevation and one angle in depression derive the distance and elevation formula. Also write formulat to find out elevation of staff staion. For given case staff reading at fix vanes are 5m and 1m, angles are 3° 45' and -2° 30'. A reading of 1.875 m was also taken on a staff held at B.M of R.L 258.5m Find horizontal distance and R.L of staff station.

Q-3 Attempt Any Two [14]

Q-3(A) Explain following terms

(i) True Value (ii) Most Probable Value (iii) Residual error (iv) Mistake (v) Systematic error (vi) Accidental error (vii) Trigonometric leveling

Q-3(B) Write laws of weight

Q-3(C) Three angles were measured closing the horizon. The values of angles are

A=102° 48'51" (Weight=3),

B=85° 42' 37" (Weight=2)

C= 171° 15' 55" (Weight=3) Find most probable value for angle A, B and C

SECTION – II

Q-4 What is remote sensing? Give difference between active remote sensing and passive remote sensing. [07]

Q-5 Attempt Any Two [14]

Q-5(A) Write detail procedure to carry out aerial surveying.

Q-5(B) Explain following terms.

(i)Altitude (ii) Flying height (iii) Isocentre (iv) Ground nadir (v) Fiducial Marks (vi) GIS (vii) GPS

Q-5(C) Calculate number of photographs (Size 250 x 250mm) required to cover an area of 20km x 16km if longitudinal overlap is 60% and side overlap is 30%. Scale of photographs is 1cm = 150m

Q-6(A) Draw the celestial sphere and highlight Astronomical triangle, also show Azimuth, Latitude, Hour angle, Declination, Vertical circle and Equator. [07]

Or

Q-6(A) Determine azimuth and altitude of star for Latitude of observer 45° N, Hour angle of star 40°, Declination of star 18° 20' N [07]

Q-6-(B) Two station A and B 100 km apart. The elevation of A is 180m and that of B is 880m. In the line of sight between A and B, there are two intervening high points C and D. C is 42 km from A and D is 81 km from A. The elevation of peaks C and D are 300m and 655m. Check whether the line of sight from A to B clears the peaks with minimum clearance of 3 m above ground level. Determine height of signal at B for intervisibility. [07]

Or

Q-6-(B) Explain procedure for extension of base line. Also write criteria for selection of base line. [07]

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